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Oak Wilt in Eastern Forests

This article describes all facets of oak wilt in eastern forests: how to identify it, manage it, and dispose of infected material.

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A red oak dying from oak wilt (Photo credit: Calvin Norman).

Oak wilt is a fungal disease that kills oaks (*Quercus* spp.) in both urban and rural settings in Pennsylvania and across the country. This article informs the reader about oak wilt, how to identify it, and treat it in forest settings. For trees in urban, backyard, and community settings, see Stay Alert with Oak Wilt!, an article by Sandy Feather.

What is Oak Wilt?

Oak wilt is a serious vascular disease caused by the fungus *Bretziella fagacearum*, formerly known as *Ceratocystis fagacearum*. It can affect oaks in both urban and rural areas. It was first reported in the United States in Wisconsin in the 1940s, and by the early 1960s, it had been confirmed in Pennsylvania. Oak wilt has been confirmed in most of Pennsylvania.

The origin of oak wilt is unclear; it may be an introduced pathogen with origins in Central America, South America, or Mexico, but it could be native to North America. Oak wilt kills trees by preventing the movement of water and sugars to the crown. This causes the leaves to wilt and die, hence the name oak wilt.

Oaks in the red oak subgenera [including but not limited to northern red oak (*Quercus rubra*), black oak (*Q. velutina*), scarlet oak (*Q. coccinea*), northern pin oak (*Quercus ellipsoidalis*), and shumard oak (*Quercus shumardii*)], chestnuts (*Castanea spp.*), and some other species are highly susceptible to oak wilt and usually die within several months of infection. The white oak subgenus [including white oak (*Quercus alba*), chestnut oak (*Q. montana*), and bur oak (Q. *macrocarpa*)] appear to be more resistant to the pathogen due to tylosises in their vascular system. These are small "bubbles" in wood fibers, which makes it easier for infected trees to contain fungus. While most infected white oaks die from oak wilt, the fungus takes much longer to kill them (up to seven years with an infection), and it has been reported that some white oaks survived infections.

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How Do I Identify Oak Wilt?

To identify oak wilt, look for sections of the crown where leaves have wilted, turned brown, or dropped. Symptoms are most easily noticed between May and August when the fungus is active, and leaves should be green and healthy. The infection typically starts with a branch or two dying back in the upper crown. As the infection spreads, the entire crown dies (Figure 1).

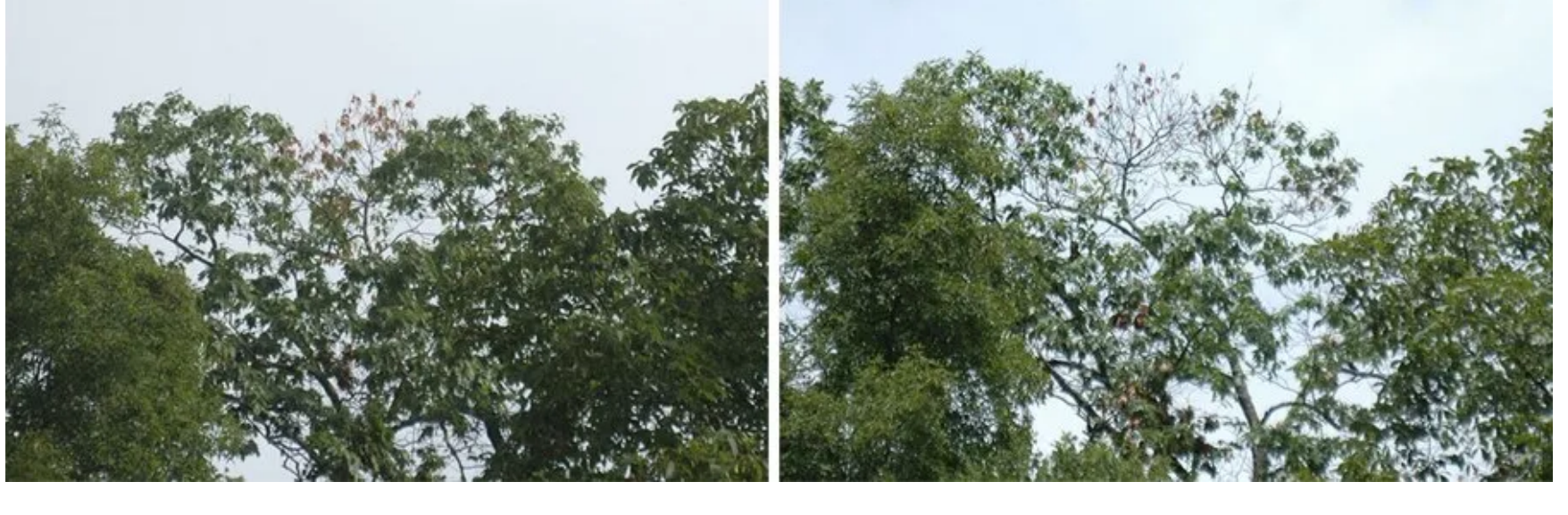


Figure 1 (left) and b (right). The progression of oak wilt in a northern red oak (*Quercus rubra*) on the Stone Valley Experimental Forest, Petersburg, PA. These pictures were taken less than a month apart. The crown of the infected tree is rapidly dying (Photo credit: Calvin Norman).

Another telltale sign of oak wilt is the distinct pattern of leaf discoloration. Leaves infected with oak wilt die from the margins inward. The last living tissue on an infected leaf is usually around the midvein at the base of the leaf (Figure 2). Dying leaves can be seen in the crowns of infected trees and found on the forest floor near symptomatic trees. Complete leaf drop usually occurs by mid- to late-summer in the more susceptible red oak species. This is much earlier than healthy trees. Dark streaking from the fungus may also be seen beneath the bark in infected branches (Figure 3).

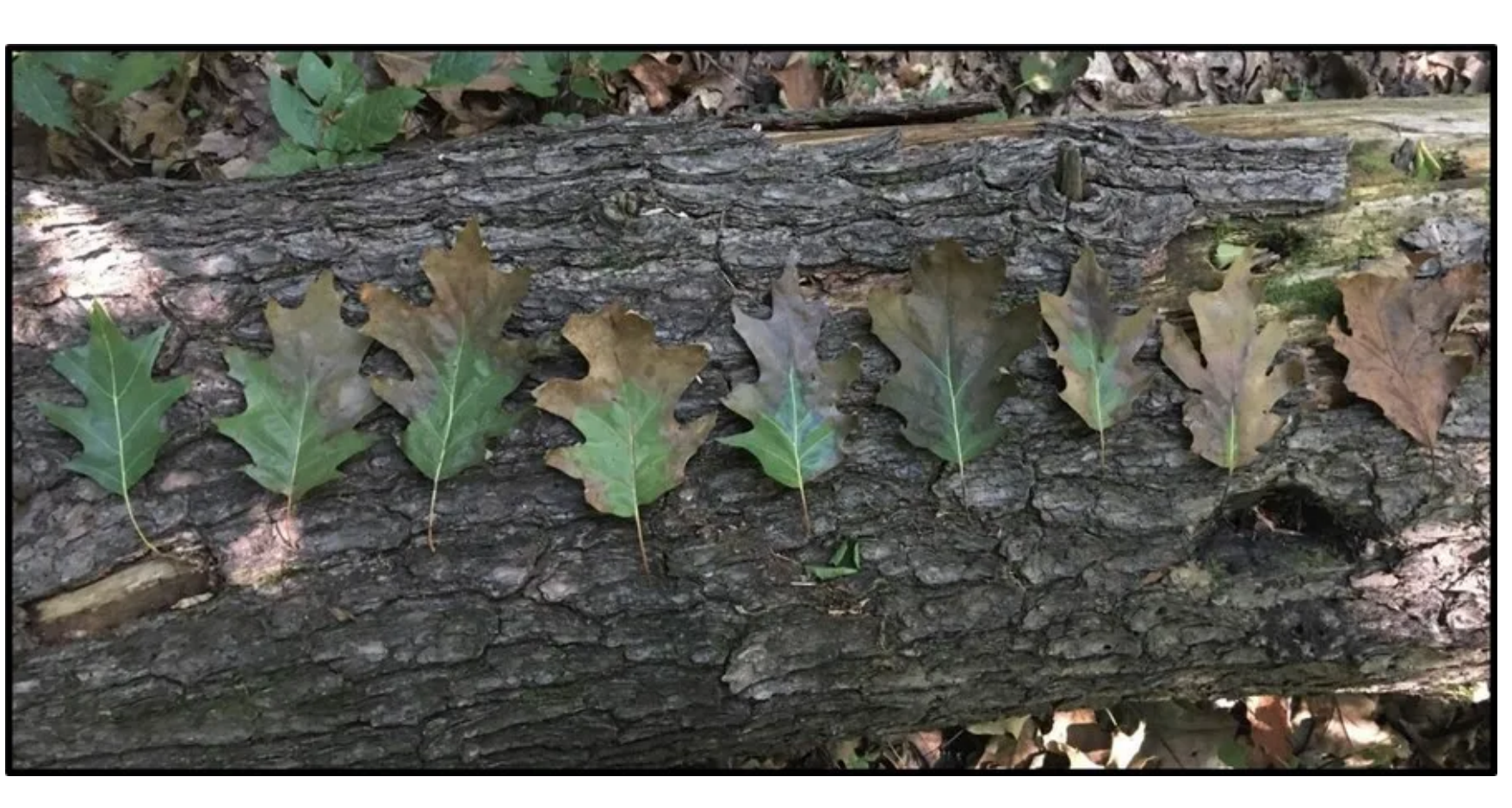


Figure 2. The stages of leaf browning due to oak wilt are shown on northern red oak leaves. Leaf tissue surrounding the midvein is the last to die (Photo credit: Calvin Norman).

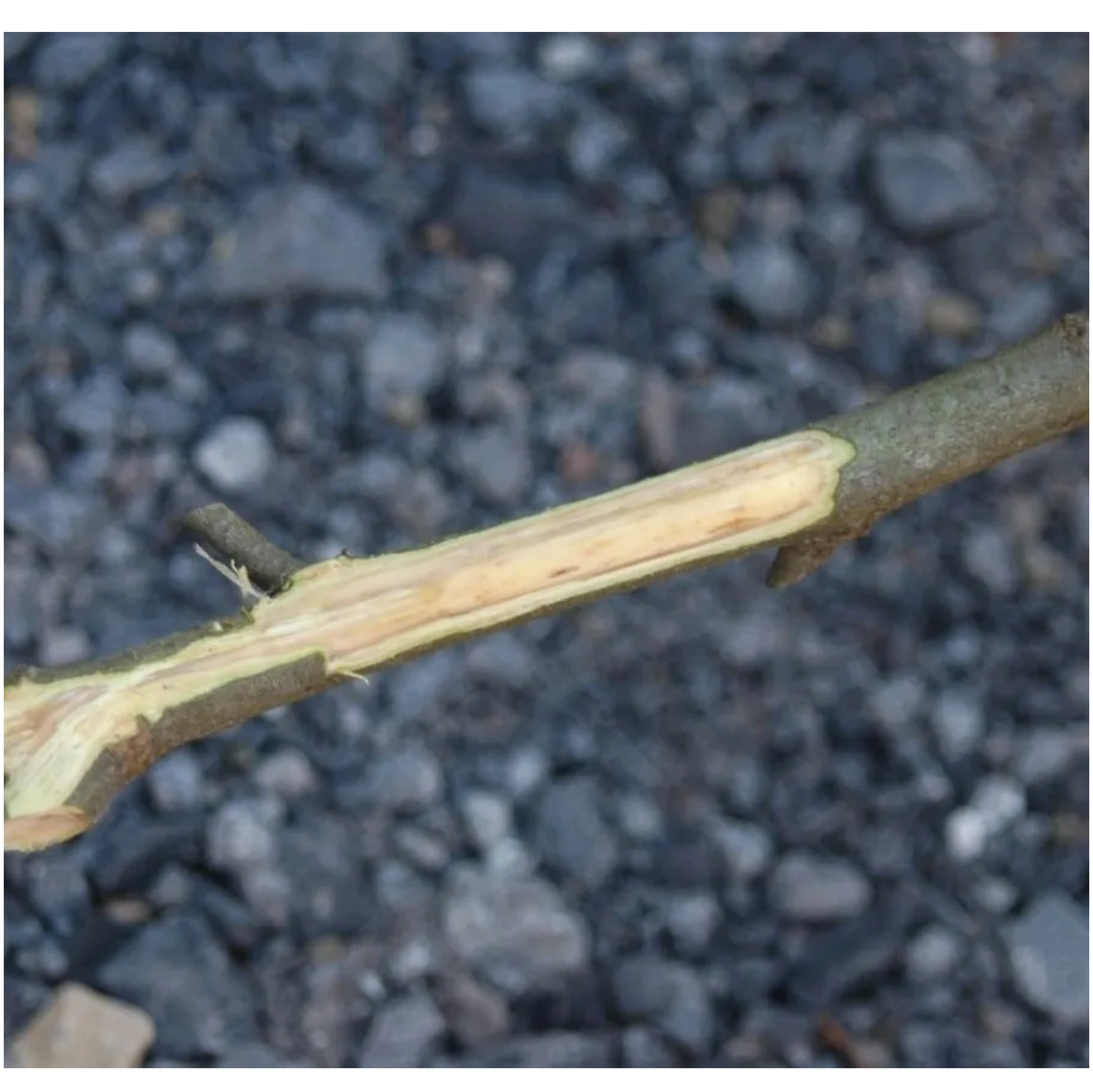


Figure 3. A branch of an infected northern red oak with dark streaks from the oak wilt fungus (Photo credit: Calvin Norman).

In the later stages of infection, typically the year after initial infection for red oaks, the fungus forms large black mats under the bark (Figure 4). These mats are called fungal spore mats or pressure pads. As they form, the bark above them cracks and splits open. Early on, they may be hard to identify, especially if they are high on the trunk. After the tree dies, sections of bark may fall off much earlier than is typical of dead oaks.

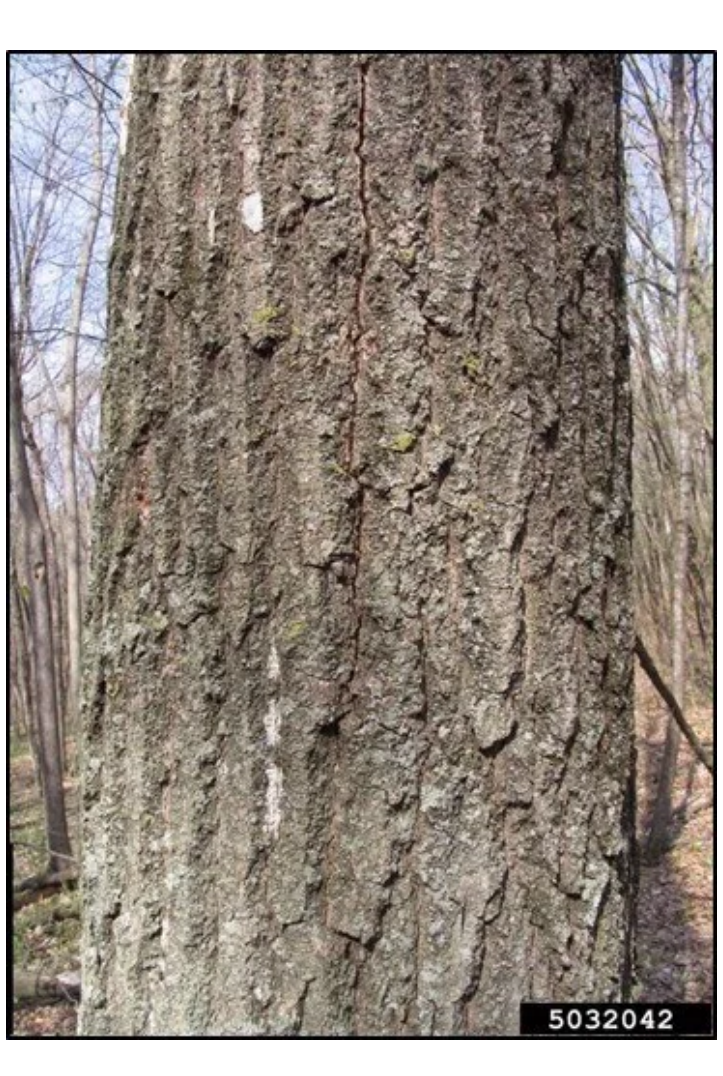


Figure 4. Cracks in the bark of a red oak indicate the presence of fungal mats (Photo Credit: Joseph Obrien).

Are There Other Diseases That Look Similar to Oak Wilt?

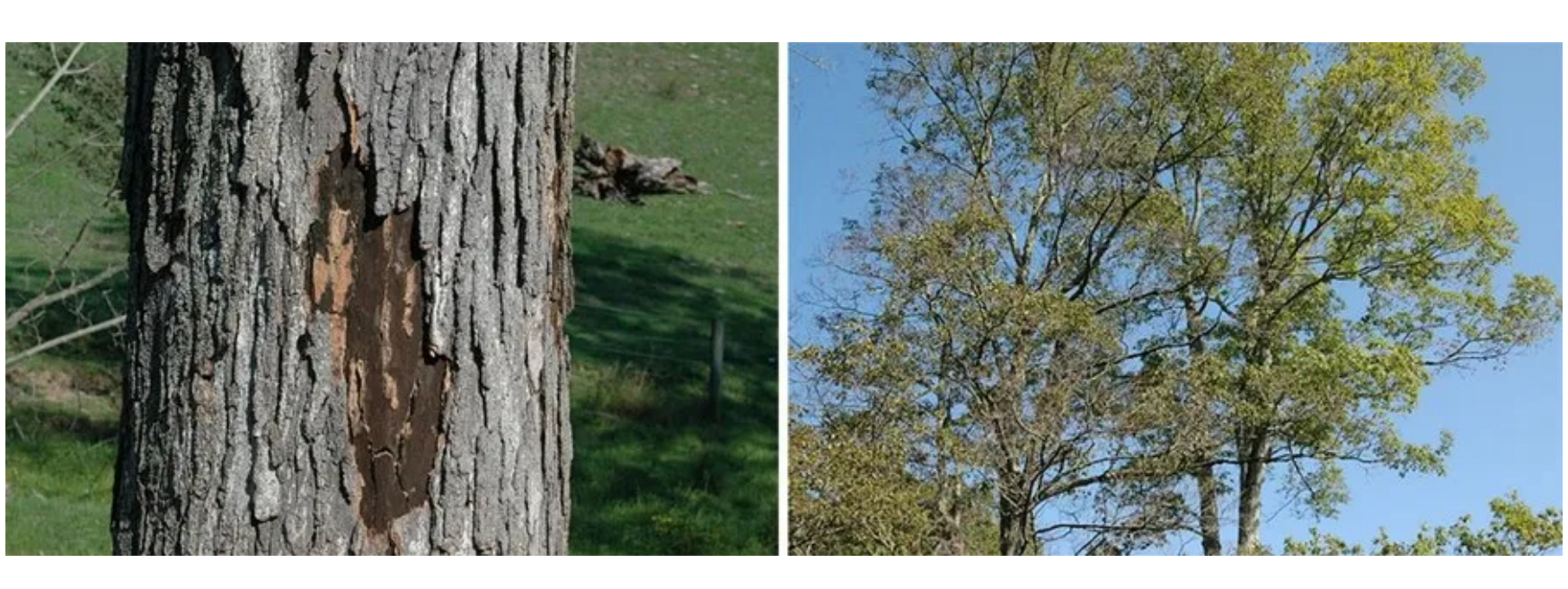


Figure 5A. Hypoxylon canker (*Hypoxylon* spp.) on the trunk of a maple (Photo credit: Calvin Norman). The brown area is the fungal body of hypoxylon canker. 5B The crown of the maple being attacked by hypoxylon (Photo credit: Calvin Norman).

Hypoxylon canker (*Hypoxylon* spp.)

Hypoxylon canker is a native opportunistic fungus that attacks deciduous trees. It is a "weak" fungus and typically just finishes off stressed trees. Infections are always fatal, but trees may take several years to die. It can be identified by the large sections of ashy black fungus on the trunk of the tree (Figure 5A), thinning canopy (Figure 5B), and unusually small leaves (leaves maybe the size of a quarter). Trees attacked by hypoxylon canker have large patches of fungus on their trunks (Figure 5A). They may sprout small leaves a year after the initial attack, and the canopy dies from the inside out. Trees attacked by oak wilt show none of these symptoms.

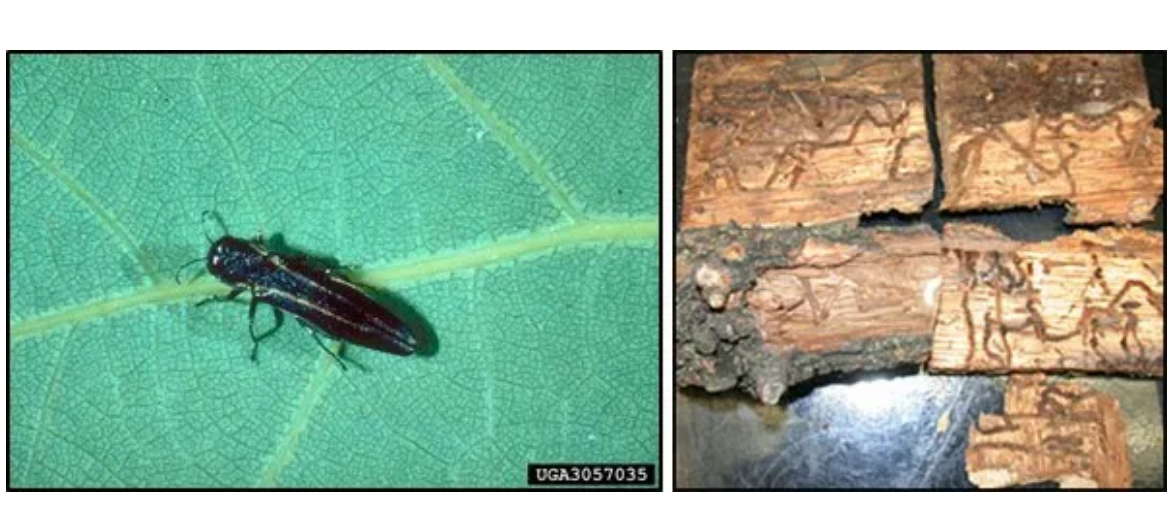


Figure 6a (left). An adult two-lined chestnut borer (*Agrilus bilineatus*; photo credit: Robert A. Haack). Figure 6b (right). Galleries carved by larval two-lined chestnut borer (Photo credit: University of Wisconsin Extension).

Two-lined chestnut borer (*Agrilus bilineatus*)

Two-lined chestnut borer is a native black beetle with two yellow stripes on its back (Figure 6A) that attacks stressed oaks, especially red oaks. Larvae do most of the damage as they burrow in the sapwood of the tree's trunk and branches; adults do little damage, as they feed on leaves. Burrowing girdles areas on the tree trunk eventually kills the tree. It takes one to five years for an attacked tree to die. Trees generally begin to display scattered branch death in late summer, as the attack progresses more branches die. Two-lined chestnut borer attacks can also be identified by peeling back the bark on branches and looking for galleries carved by burrowing larvae (Figure 6B) or a "D" shaped-hole larval carve as they emerge.

How Does Oak Wilt Spread?

Oak wilt spreads via three major pathways: overland, underground, and human-mediated.

Overland spread is mediated by Nitidulidae beetles—a family of native sapsucking insects—that feed on sap, particularly from injured trees. These beetles are attracted to trees by the smell that oaks release after being injured. Additionally, these beetles are also attracted to the sweet smell of fungal. They spread oak wilt when they visit a spore mat and pick up spores, and then visit an uninfected tree. Overland spread usually occurs after damaging events like strong storms or logging, these events cause canopy damage, which then attract Nitidulid beetles.

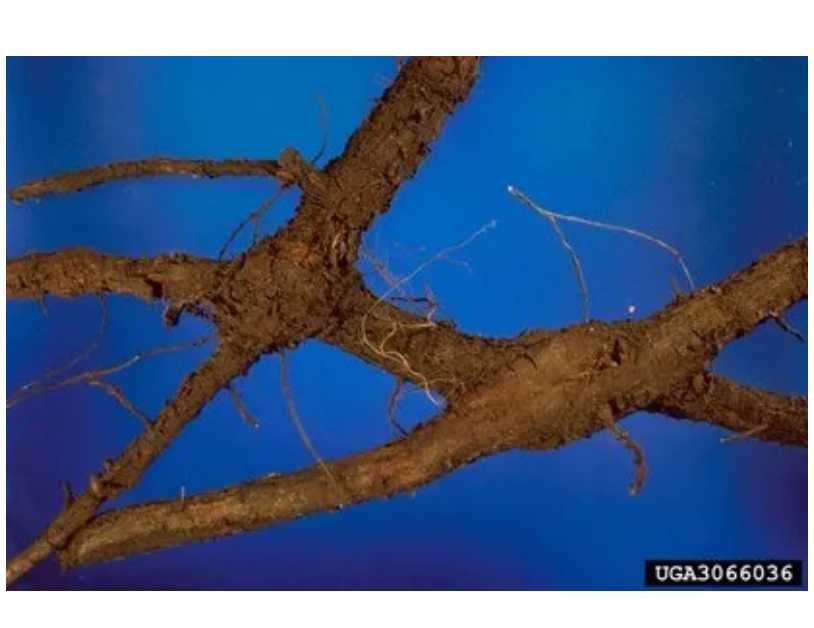


Figure 7. Two oaks connected by root grafts, which is how underground spread of oak wilt occurs. (Photo credit: James Solomon, USDA Forest Service, Bugwood.org)

Underground spread of oak wilt takes advantage of oak physiology. Oak roots often graft to the root systems of nearby oaks of the same species—this creates an interconnected network underground (Figure 7). Trees use these grafts to share resources in times of stress, like nutrients and water. In areas with the right soil conditions and topography, acres of oaks can be interconnected through grafted roots.

Oak wilt uses these grafts to spread. As oak wilt spreads within an infected tree, it travels throughout the tree's vascular tissue, which extends from the uppermost branch to the smallest root. Once it reaches the root system, oak wilt can travel through the roots of an infected oak to any oaks that are grafted to the infected tree's roots. This can result in one infected tree leading to large pockets of infected oaks.

In the Midwest, where soil conditions are very conducive to root grafting, untreated oak wilt cases can kill acres of oak trees. To identify underground spread, look for clusters of dying oaks of the same species that do not have crown injuries or other obvious signs of stress. The most advanced symptoms of oak wilt will be apparent in the center of the outbreak, likely where oak wilt entered the stand, with less developed symptoms at the margins of an affected cluster.

It is believed that equipment (e.g., chainsaws, pruning saws, or tree climbing spikes) can also be a vector for oak wilt. For example, a pruning saw, used on an infected tree could spread infected sawdust off-site or introduce fungal spores into new wounds on uninfected trees. In this way, infected equipment can act like Nitidulid beetles. This type of infection could be common along rights-of-ways or in any landscapes where the same equipment is used on multiple trees.

The fungus can survive in untreated infected material (e.g., logs) for up to a growing season. Using unsanitized equipment and moving infected material from one site to another allow oak wilt to be moved over long distances (Figure 8). Another major source of human-mediated spread is the transfer of infected tissue from one place to another (for example, moving firewood).

Figure 8. An arborist sanitizing pruning saw after collecting an oak wilt sample (Photo credit: Michael Houtz).

Managing Oak Wilt at Stone Valley

Here is an example of how an oak wilt outbreak may progress: *On June 21, a windstorm broke a large branch off a mature northern red oak along the edge of a woodlot. That day, Nitidulid beetles that had been congregating on a nearby oak wilt fungal mat fly toward the smell of fresh sap in the new wound, carrying with them oak wilt spores. This infects the damaged tree. The initial outbreak went unnoticed and unmanaged. By September 15, the newly infected oak had lost all its leaves, and the fungus had already spread by root grafts to four nearby red oaks. By the following summer, the fungus had advanced through oaks grafted to the infected tree, and fungal spore mats had developed for sapsucking insects to find and move to new areas, continuing to spread oak wilt. This cycle may continue until the outbreak is treated or until all surrounding oaks are killed.*

How Do I Confirm or Test for Oak Wilt?

Identifying oak wilt in mature forests can be difficult, but if it is detected early and responded to rapidly, large-scale outbreaks can be avoided. The numerous trees and closed canopy make it challenging to find and fully examine infected trees. The best way to identify oak wilt in forests is to look for all the symptoms of oak wilt: symptomatic leaves, crown dieback, and dark streaks under the bark of infected twigs (Figure 3).

Identification can be difficult after mid-September as other factors, such as early leaf fall due to stress and other forest pests, can be easily confused with oak wilt. The easiest way to find oak wilt is to examine suspected areas for thinning crowns from an open area like a road or clearing or from above (Figure 1). It can be difficult to determine symptoms of infection in tall mature trees with just the naked eye, but a pair of binoculars will help.

While at the tree, look on the forest floor for symptomatic leaves (Figure 2). It is best to only use freshly fallen leaves that are near the trunk. Low branches or stump sprouts can be helpful in identifying oak wilt; strip the bark and examine the wood for dark streaks (Figure 4). Drones can be helpful in scouting for infected trees.

Key Steps for Oak Wilt Identification in Forest Settings

- Look for thinning crowns and rapid defoliation, within a few weeks
- Look for symptomatic leaves at the base of the tree that have browned from the outside margins inward to the base
- Strip bark from low twigs or stump sprouts and look for dark streaking
- Look for dark fungal pressure pads under cracked bark on the stem in older infections
- Look for clusters of trees with dead oaks at the center and symptomatic trees at the margins
- Suspected infections should be confirmed with laboratory testing.

If you suspect an oak wilt infection, the tree(s) should be laboratory tested to ensure oak wilt is the damaging agent. In Pennsylvania, the Penn State Plant Disease Clinic is a good option. To take a sample, collect sections from several symptomatic branches (0.5-1.0 inch in diameter and 6 inches long). Place the samples in a labeled plastic bag and then put them into a cooler right away (Figure 9). The fungus is very heat-sensitive! For example, samples of the oak wilt are known to die in hot cars. Be sure to take care of any samples to prevent a false negative test. Due to their size, mature trees with tall crowns may be difficult to sample. In these cases, use a pole saw to sample the lowest branch or hire an arborist to climb the tree and obtain a sample.

As oak wilt is present throughout the vasculature of the entire tree, stump sprouts and low branches can be used if available. Be sure to sanitize all equipment before and after collecting each sample. This is done by wiping equipment down with sanitizing wipes or by spraying and wiping down equipment with a 10% bleach solution.

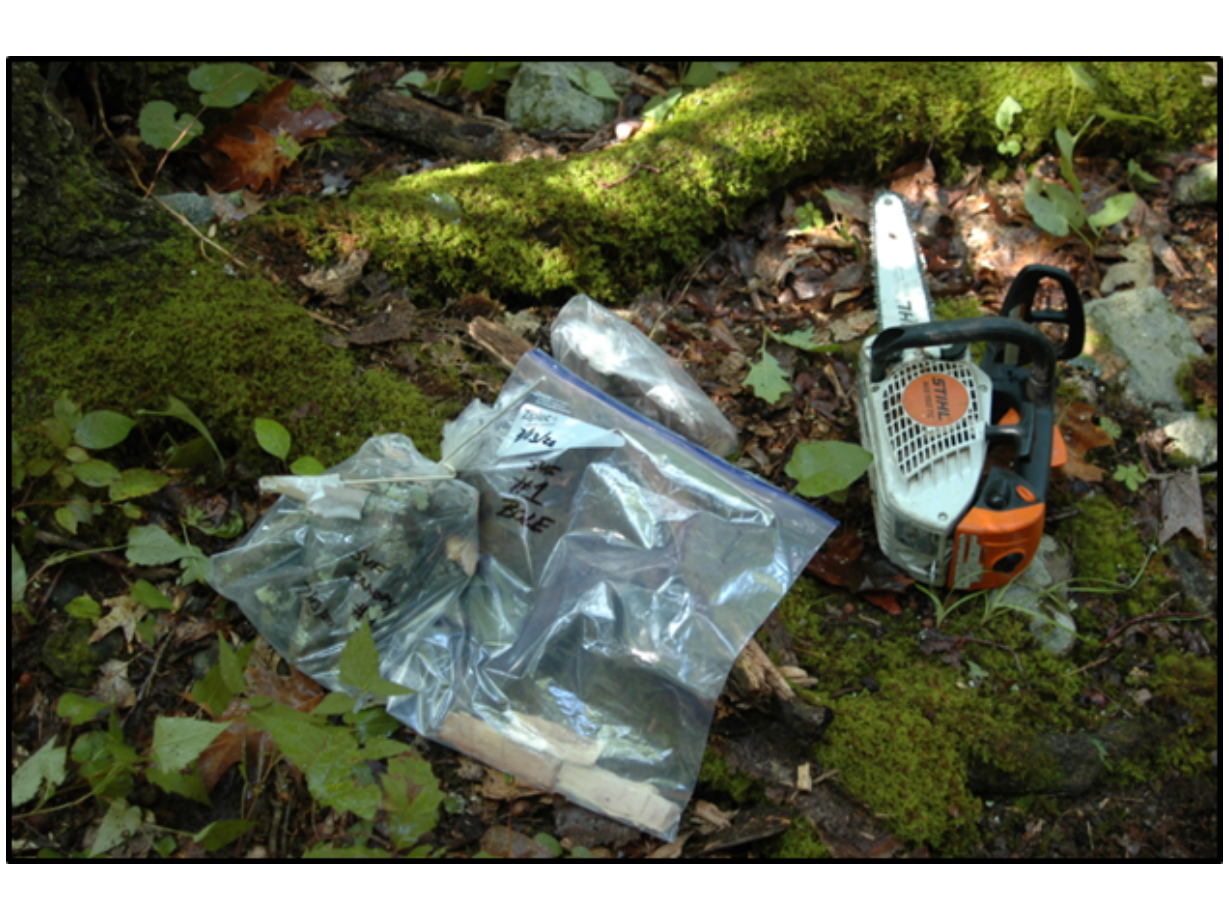


Figure 9. Samples collected from oaks ready for testing at the Penn State Plant Disease Clinic (Photo credit: Michael Houtz).

If collecting a branch sample is not possible, a sample from the trunk may be taken. Use a hatchet or ax to cut into the inner bark of the tree. Then use a chisel to collect some of the sapwood (Figure 10). This type of sampling is a last resort, as it is very invasive and may increase a tree's susceptibility to oak wilt and other diseases. It should only be used in cases where oak wilt is very likely the cause of the wilting and defoliation and a branch cannot be collected. Be sure to sanitize all equipment following collection.

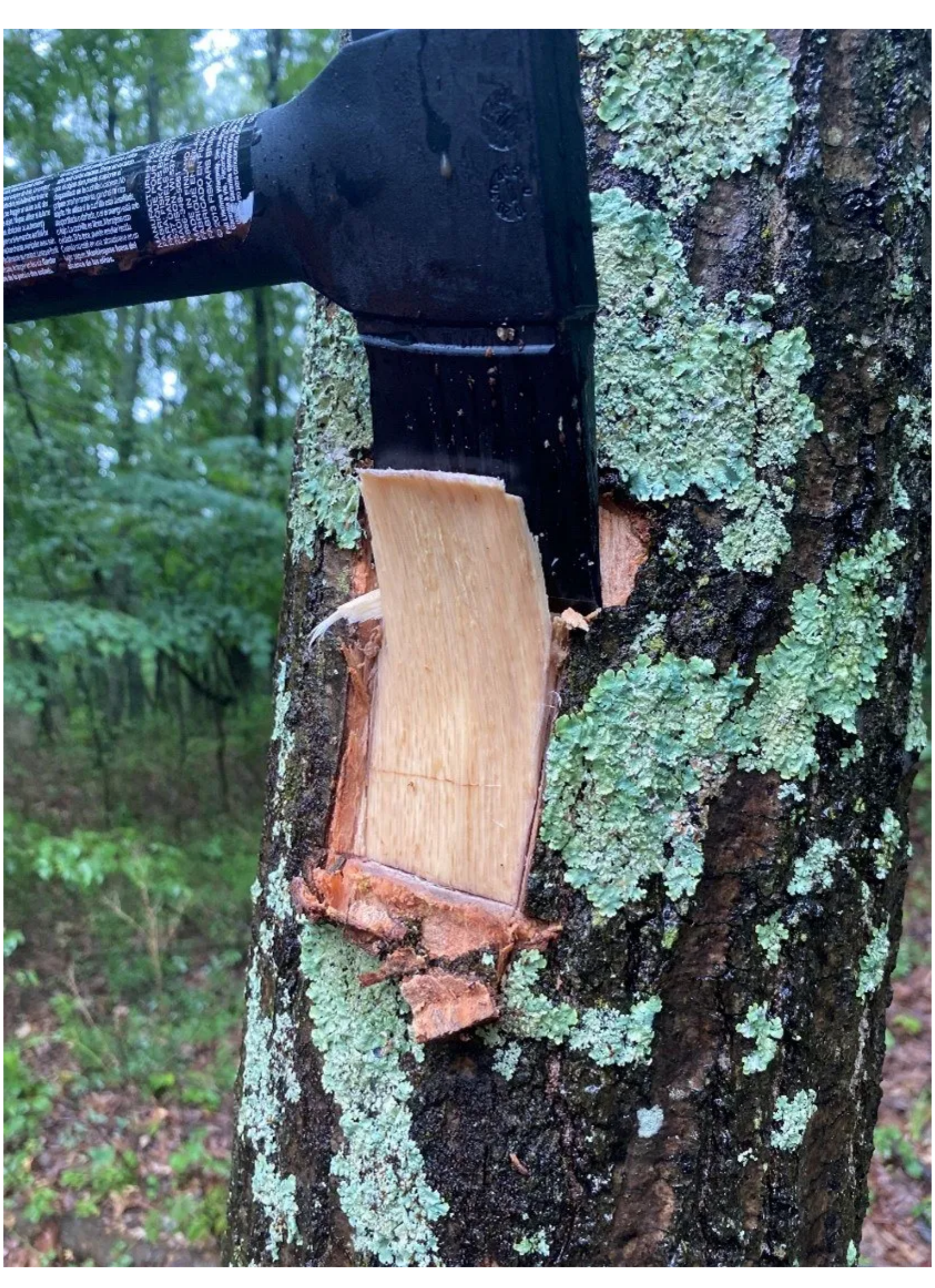


Figure 10. Collecting a trunk sample from an oak tree with suspected oak wilt (Photo credit: Sarah Wurzbacher).

What are My Management Options?

In urban and community forests and in forests in other regions, typical management response to an oak wilt infection involves intensive trenching and root plowing to disrupt root grafts and halt underground spread in a buffer around infected trees. In small pockets of oak wilt, a 100-foot buffer is used to ensure all infected or potentially infected trees are captured, but more precise buffer distances may be calculated based on the infected tree's diameter as well as the soil type and topography. For more information about buffer distances, refer to the Minnesota Department of Natural Resources Oak Wilt Management Guide. At the buffer perimeter, all tree roots need to be cut with a root plow to prevent further underground spread. Root cutting, or trenching, should occur soon after the infection is discovered. In addition, during the dormant season (October-March) the infected oak, and any surrounding oak that fall within the buffer, need to be cut, salvaged, and removed. All infected trees need to be cut down and properly disposed of to prevent future spread. Though effective, this approach involving root plowing can sometimes be financially and logistically prohibitive in forest settings (Figure 11).

An alternative management strategy can be used to manage oak wilt—infected trees, and nearby trees that may be infected are killed using triclopyr. Dead trees can be removed or left on site as wildlife habitat. This strategy disrupts the root connections around infected trees to prevent underground spread. It has been successfully used by Penn State Extension and the DCNR. For more information, read the Minnesota Department of Natural Resources Oak Wilt Management Guide.

Additionally, time operations to avoid the riskiest infection windows (May to mid-July), process wood to encourage drying (which renders the fungus nonviable), and properly dispose of infected material.

How is Infected Wood Handled?

All trees that are 10 inches in diameter and greater may not dry sufficiently in one season and are, therefore, a potential reservoir for future infection. They must be properly processed or disposed of before April of the year following initial infection to prevent insects from picking up spores and moving them to new infection sites. Infected wood can be disposed of in several ways.

- Sawlogs and pulpwood may be utilized from infected trees
- Potential buyers must be notified that logs are infected; logs for export may require additional processing or treatment to kill any fungus (e.g. fumigation, steam, and heat treatments)
- Sawlogs and pulpwood should be debarked by spring
- Whole trees or residues may be chipped to encourage dryingWhole trees or residues may be incinerated
- Firewood can be harvested from infected trees, but it must be cut, split, stacked, and dried by the following spring
- Firewood should not be moved from the infection site and should be burned as soon as possible.

See Figure 11 for a management calendar.

Preventing Oak Wilt

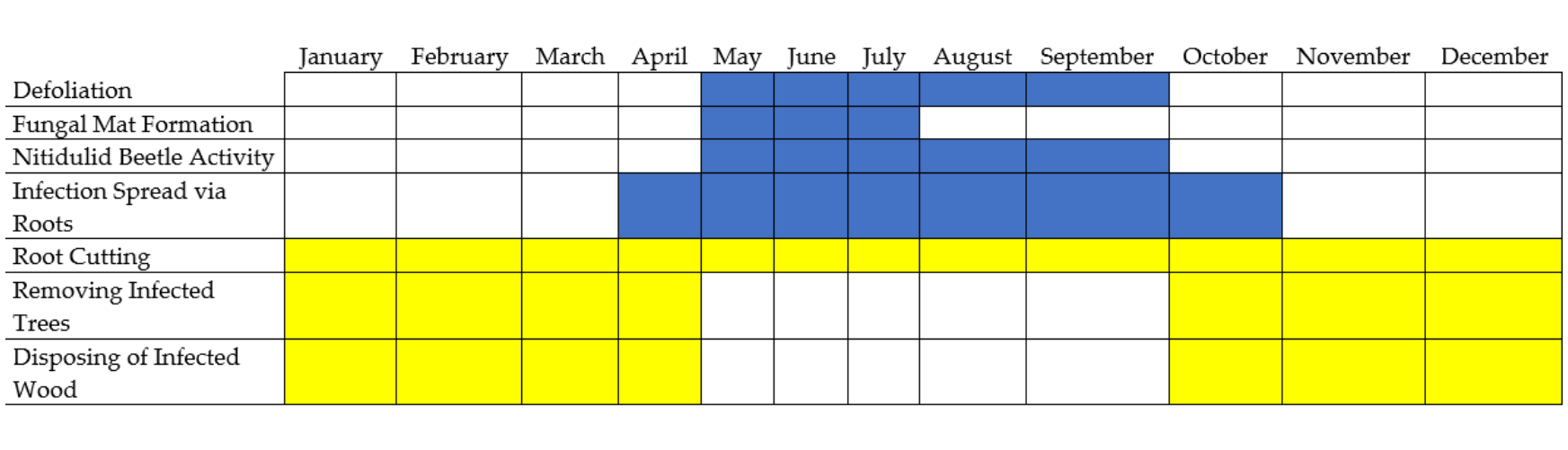
When dealing with oak wilt, an ounce of prevention is worth a pound of cure. All cutting equipment, such as pruning saws and chainsaws, needs to be properly sanitized between uses. To clean equipment, first brush away sawdust and wood chips and then spray with a 10% bleach in water solution (Figure 8). Sawdust and chips from infected trees can carry the fungus, so be sure to clean equipment either on site or distanced from trees, like in a garage.

When possible, do not injure oaks through pruning or logging activities. For example, when oak wilt fungus is most active (April to at least July but ideally October). Harvesting timber where oak wilt is an issue would best be done during the winter months, as harvesting may damage residual trees and create new infection sites. It is also important not to move "green" firewood. Wood that is not properly dried may spread oak wilt.

Conclusion

While managing oak wilt seems daunting, it can be done. Successful management requires rapid identification and response, followed by the proper disposal of infected material. Doing so will reduce the chance of oak wilt spreading and impacting nearby oaks. Oak wilt can be prevented by implementing proper management, sanitation, and disposal practices.

Oak Wilt Management Calendar



	January	February	March	April	May	June	July	August	September	October	November	December
Defoliation					🟦	🟦	🟦	🟦	🟦			
Fungal Mat Formation					🟦	🟦	🟦					
Nitidulid Beetle Activity					🟦	🟦	🟦	🟦	🟦			
Infection Spread via Roots				🟦	🟦	🟦	🟦	🟦	🟦	🟦		
Root Cutting	🟨	🟨	🟨	🟨	🟨	🟨	🟨	🟨	🟨	🟨	🟨	🟨
Removing Infected Trees	🟨	🟨	🟨	🟨						🟨	🟨	🟨
Disposing of Infected Wood	🟨	🟨	🟨	🟨						🟨	🟨	🟨

Figure 11. Logging activities that could damage neighboring oaks, pruning and cutting activities in areas at high risk for oak wilt, and hauling infected material should be restricted to times of the year when the fungus is not active. Disposal can be conducted at any time, but infected material must be disposed of by April.

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